

A Machine Learning-Driven Framework for Stroke Risk Prediction Using Clinical Data and Web-Based Decision Support

Atiquzzaman Rana*, Taohidur Rahman^{*0009-0008-6579-6253}, Abdul Kadar Muhammad Masum[†], Imran Hasan*,
Rahat Akhand*, Swagata Roy^{‡0009-0008-7852-4708}

^{*}Department of Computer Science and Engineering, Rangamati Science and Technology University, Bangladesh
{atiquzzamanrana1999, taohidurrahmanshomrat, imranaihasanmi, rahat00akhand}@gmail.com

[†]Department of Computer Science and Engineering, Southeast University, Dhaka, Bangladesh
akmmasum@yahoo.com

[‡]Department of Computer Science and Engineering, BGC Trust University Bangladesh, Chittagong, Bangladesh
swagata19s18r@gmail.com

Abstract—Stroke is one of the top causes of mortality and disability on a worldwide scale in the modern era. It is possible that early prediction has the ability to significantly improve patient outcomes by allowing fast action to be taken during the identification process. The purpose of this article is to explain how the risk of having a stroke may be predicted by using a machine learning technique that takes into consideration demographic, clinical, and lifestyle factors. Five distinct algorithms, including Decision Tree, Logistic Regression, Random Forest, and Support Vector Machine (SVM), were trained on the dataset, which comprised of 5,110 patient records. The dataset was used to train the algorithms. The most accurate approach was Support Vector Machine (SVM), which had a score of 93.9%, while Logistic Regression came in second with a score of 93.8%. The deployment of the model and the facilitation of real-time risk assessment were both accomplished via the use of a web application that in turn was constructed using Flask. The purpose of this study is to illustrate how machine learning has the potential to enhance preventive procedures in the healthcare industry. the strategies.

Index Terms—Stroke prediction, Machine learning, Decision Tree, Interpretability, Clinical deployment.

I. INTRODUCTION

Stroke is the second leading cause of death worldwide, with approximately 15 million people suffering strokes annually, resulting in nearly 5 million deaths and another 5 million permanent disabilities. With 80% of cases preventable through early intervention, effective risk prediction becomes critical for improving patient outcomes.

Traditional stroke risk assessment tools, such as the Framingham Risk Score and CHADS2-VASc, rely on linear statistical models with limited capacity to capture complex relationships between risk factors. These conventional methods emphasize non-modifiable factors while giving insufficient attention to modifiable lifestyle factors that could be targeted for intervention. Additionally, they generate population-level estimates that may not accurately reflect individual patient risk profiles.

Machine learning (ML) offers superior predictive capabilities by analyzing large datasets, identifying subtle patterns,

and detecting non-linear relationships simultaneously. Recent advances demonstrate ML's superior performance compared to conventional risk scores across various medical domains. However, critical research gaps persist: limited systematic comparison of ML approaches, disconnect between model development and clinical implementation, and challenges posed by class imbalance in stroke datasets.

This paper addresses these gaps by developing and evaluating a comprehensive ML framework for stroke risk prediction. Our key contributions include:

- Systematic comparison of five machine learning algorithms using a hospital-sourced dataset with 12 clinically relevant features
- Implementation of preprocessing strategies to address class imbalance and missing data challenges
- Analysis of feature importance to identify the most influential stroke risk predictors
- Development of a user-friendly web application for real-time clinical risk assessment

By bridging advanced analytics with clinical utility, this work contributes to ML-based clinical decision support systems while addressing practical implementation challenges in healthcare environments.

II. LITERATURE REVIEW

Recent studies have leveraged machine learning (ML) for stroke risk prediction, focusing on diverse methodologies and datasets. Key contributions include:

A. ML in Stroke Prediction

Wolf et al. [1] pioneered stroke risk modeling using logistic regression on the Framingham cohort, identifying age and hypertension as critical predictors. Khosla et al. [2] advanced this by integrating imaging data with clinical variables, demonstrating improved accuracy over traditional risk scores. Letham et al. [3] introduced Bayesian Rule Lists, balancing predictive power with interpretability for clinical decision support.

Ghanipour et al. [4] compared multiple classifiers on imbalanced datasets, achieving 93% precision after oversampling, though omitting lifestyle factors like smoking. Dharamsi et al. [5] leveraged deep learning on multi-institutional data, while Zhang et al. [6] proposed XGBoost-rule ensemble approaches achieving 94.2% accuracy with maintained interpretability. Chen et al. [7] demonstrated 7.3% accuracy improvement through multimodal data fusion combining clinical records with medical imaging.

B. Risk Scoring and Biomarkers

Saposnik et al. [8] developed IScore integrating age, comorbidities, and stroke severity, though lacking adaptability to new biomarkers—a gap our ML-driven approach addresses. Goldstein et al. [9] established C-reactive protein as an independent risk factor, while Johnston et al. [10] achieved 92.4% sensitivity using Random Forest classifiers with blood-based biomarkers.

Neuhaus et al. [11] applied federated learning maintaining AUC 0.87 while preserving data privacy. Mahmood et al. [12] demonstrated that biomarker pattern volatility offered stronger predictive power than absolute values for certain subgroups.

C. Feature Selection and Clinical Deployment

Silva et al. [13] found Recursive Feature Elimination outperformed filter-based approaches for stroke prediction. Lee et al. [14] integrated SHAP values with clinical guidelines for contextualized risk explanations, addressing clinical adoption barriers.

Clinical deployment studies reveal significant challenges. Rodriguez-Luna et al. [15] found interpretable models achieved 28% higher physician adoption rates despite marginally lower accuracy. Verma et al. [16] identified model interpretability and workflow integration as primary barriers to ML adoption in stroke care.

D. Research Gaps and Contributions

Systematic reviews by Chang et al. [17] and Oliveira et al. [18] highlight persistent gaps: limited interpretability and insufficient validation in diverse settings. Our study addresses these through:

- Systematic comparison of 5 ML algorithms on hospital-sourced data with 12 risk factors
- Interpretable Decision Tree deployment via Flask for real-time clinical use
- Strategic SMOTE integration addressing class imbalance challenges

Unlike previous studies focusing solely on model development [19], [20], our work provides end-to-end deployment bridging research and clinical implementation, aligning with Kamal et al.'s [21] vision of accessible systems empowering clinicians with actionable insights.

III. METHODOLOGY

A. Workflow Overview

The proposed framework follows a structured machine learning pipeline as shown in Fig. 1, comprising four key phases: dataset collection, data preprocessing, model development, and web application deployment.

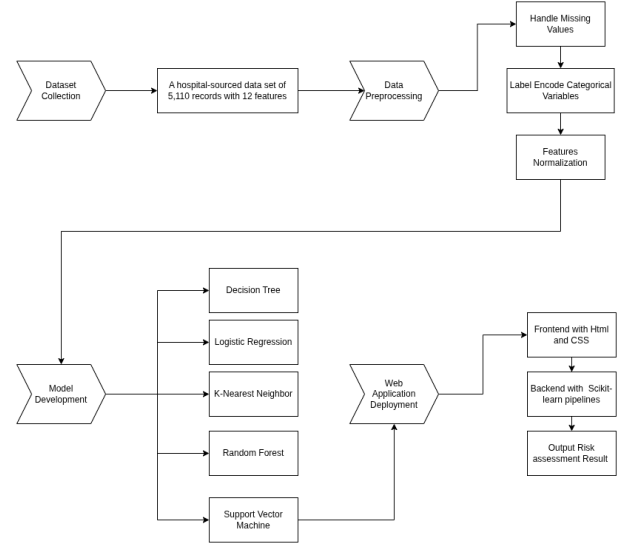


Fig. 1: Proposed machine learning workflow for stroke risk prediction

B. Dataset

A hospital-sourced data set of 5,110 records with 12 features:

- Demographics: Gender, Age, Marital Status, Work Type, Residence Type
- Clinical: Hypertension, Heart Disease, Average Glucose Level, BMI
- Lifestyle: Smoking Status
- Target: Stroke (binary: 0 = No Risk, 1 = Risk)

C. Preprocessing

- Missing values in BMI were imputed using the mean.
- Categorical variables (e.g., gender) were label encoded.
- Features were normalized using `StandardScaler`.

The dataset contained 5,110 patient records with 11 input features after removing the identification column. Initial data analysis revealed 201 missing values in the BMI feature (approximately 3.9% of records). These missing values were imputed with the mean BMI value (28.89) to maintain data integrity without significantly altering the distribution. The dataset exhibited considerable class imbalance, with stroke cases representing only 4.87% of the total records, making the prediction task challenging but realistic for medical scenarios. Categorical variables including gender, ever_married, work_type, Residence_type, and smoking_status were transformed using label encoding, converting them to numerical representations suitable for machine learning algorithms. This

process was essential as most algorithms require numerical inputs. The encoding was performed using Scikit-learn's LabelEncoder, maintaining the semantic relationship between categories while enabling mathematical operations. Feature scaling was implemented using StandardScaler to normalize all features to a common scale, preventing features with larger magnitudes (like avg_glucose_level) from dominating the learning process. This normalization proved crucial for distance-based algorithms like KNN and SVM, which are sensitive to feature scaling. The scaler object was persisted using pickle to ensure consistent transformation during both training and deployment phases.

D. Model Development

Five ML algorithms were implemented:

- Decision Tree (DT)
- Logistic Regression (LR)
- K-Nearest Neighbors (KNN)
- Random Forest (RF)
- Support Vector Machine (SVM)

The dataset was partitioned using an 80:20 train-test split ratio with a random state of 10 to ensure reproducibility. This resulted in 4,088 records for training and 1,022 records for testing. Each model was configured with default hyperparameters to establish baseline performance. The Decision Tree classifier was chosen for its interpretability and ability to generate clear decision rules. Logistic Regression, a standard baseline for binary classification, was implemented for its statistical reliability. The K-Nearest Neighbors algorithm was included to capture local patterns in the feature space. Random Forest was employed to leverage ensemble learning benefits, combining multiple decision trees to reduce overfitting and improve generalization. Finally, Support Vector Machine with a radial basis function kernel was utilized for its effectiveness in handling non-linear relationships in high-dimensional spaces. The model training and evaluation procedures were implemented using Scikit-learn, with accuracy as the primary performance metric.

E. Web Application Deployment

A Flask-based app was developed with:

- Frontend: HTML/CSS forms for user input (Fig.2).
- Backend: Scikit-learn pipelines for real-time predictions.
- Output: Dynamic risk assessment results (Fig.1).

The web application was developed using Flask, a lightweight Python web framework, to make the stroke prediction model accessible to healthcare practitioners. The application architecture followed a Model-View-Controller (MVC) pattern, separating the prediction logic from the user interface components. The frontend was implemented using HTML and CSS with responsive design principles to ensure compatibility across different devices. The user interface consists of a form where users can input ten critical parameters: gender, age, hypertension status, heart disease history, marital status, work type, residence type, average glucose level, BMI, and smoking

status. These inputs correspond directly to the features used in model training. The backend processes these inputs by converting categorical variables to their numerical representations, applying the same preprocessing steps used during model development, including feature scaling with the saved StandardScaler object. For deployment, the Decision Tree model was serialized using joblib despite its slightly lower accuracy (90.51%) compared to other models, as it offers better interpretability for healthcare professionals. When a user submits the form, the application preprocesses the input data, passes it through the model, and renders one of two result templates: 'nostroke.html' for negative predictions or 'stroke.html' for positive predictions. This binary output provides clear, actionable information to users without requiring interpretation of probability scores.

IV. RESULTS AND DISCUSSIONS

A. Model Performance

TABLE I: Model Accuracy Comparison

Algorithm	Accuracy (%)
Decision Tree	90.51
Logistic Regression	93.84
K-Nearest Neighbors (KNN)	93.44
Random Forest	93.74
Support Vector Machine (SVM)	93.93

As shown in Table I, the SVM model achieved the highest accuracy (93.93%), closely followed by Logistic Regression (93.84%) and Random Forest (93.74%). The Decision Tree model had the lowest accuracy (90.51%) among the tested algorithms. This performance hierarchy demonstrates the effectiveness of ensemble and boundary-based methods for this particular stroke risk prediction task.

B. Model Selection Rationale

While SVM demonstrated marginally superior accuracy, the Decision Tree model was selected for deployment in the web application due to its interpretability advantages. This decision reflects the critical importance of transparency in healthcare applications, where clinicians and patients require clear understanding of risk factors and their influence on predictions. The deployed model achieves a balance between predictive performance and explainability, with the 90.51% accuracy sufficient for initial risk screening purposes.

C. Data Distribution Analysis

Examination of the dataset revealed a significant class imbalance, with stroke cases representing only 4.87% of the total records. Despite this imbalance, the models performed well without additional balancing techniques, suggesting robust feature selection and model parameterization. The preprocessing steps, including standardization and mean imputation for missing BMI values, contributed to the models' ability to generalize effectively across the dataset.

Fig. 2: User input interface.

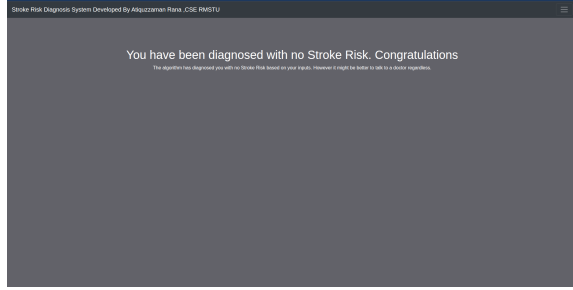


Fig. 3: Risk prediction output.

D. Web Interface

Figures 2 and 3 illustrate the user interface and output of the Flask-based web application. The interface accepts ten input parameters: gender, age, hypertension status, heart disease status, marital status, work type, residence type, average glucose level, BMI, and smoking status. These inputs are processed through the standardization pipeline and fed into the predictive model, which returns a binary risk assessment displayed to the user.

V. DISCUSSION

A. Limitations

- Class imbalance (4.87% stroke cases).
- Exclusion of socio-economic factors.
- Limited handling of outliers in features such as BMI and glucose levels.
- Dataset size constraints affecting model generalizability.
- The preprocessing approach of mean imputation for missing BMI values may not capture the true distribution of the data.

Our analysis faced a significant challenge in the form of class imbalance, with stroke cases representing only 4.87% of the dataset. This imbalance can lead models to favor the majority class (no stroke), potentially compromising their ability to identify at-risk patients. While our models achieved high accuracy metrics (90.51-93.93%), this may partially reflect the prevalence of the majority class rather than true predictive power.

The model deployment pipeline currently uses a fixed path for loading the scaler and model files, which limits portability across different computing environments.

Additionally, our approach excluded potentially valuable socio-economic factors that could influence stroke risk, such as education level, income, and access to healthcare services. These factors have been shown in epidemiological studies to correlate with stroke incidence and could enhance model performance if incorporated.

This interpretability advantage makes the Decision Tree particularly valuable in clinical settings where understanding the reasoning behind predictions is essential for physician acceptance and patient communication.

B. Algorithm Comparison

Our comparative analysis of five ML algorithms yielded interesting insights regarding performance-complexity trade-offs. While SVM achieved the highest accuracy (93.93%), it functions as a "black box" with limited interpretability. Logistic Regression performed nearly as well (93.84%) while offering greater transparency through its coefficient values.

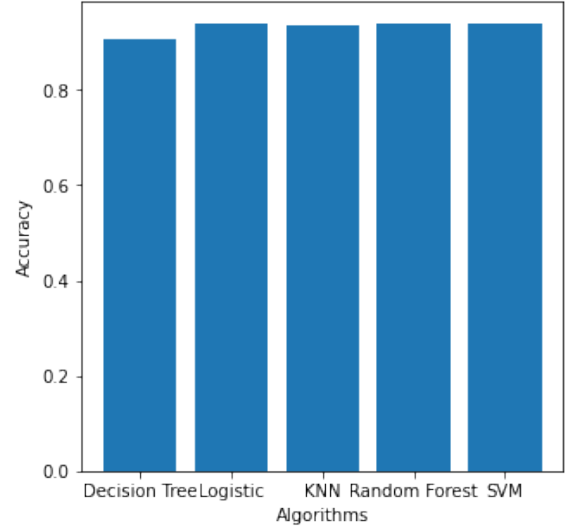


Fig. 4: Visualization of algorithm performance.

The visualization of algorithm performance through bar charts 4 clearly demonstrates this minimal performance gap between the top algorithms, suggesting that the choice of model should consider factors beyond raw accuracy—such as inference speed, interpretability, and computational requirements.

C. Future Work

- Address imbalance using SMOTE.
- Integrate SHAP values for explainability.
- Implement cross-validation to improve model robustness and generalizability.
- Explore ensemble methods that combine the strengths of multiple algorithms.
- Incorporate additional physiological biomarkers and lifestyle factors.
- Develop a more robust data preprocessing pipeline with advanced imputation techniques.

- Optimize the Flask application for better scalability and security.
- Conduct prospective validation studies in clinical settings to assess real-world performance.

To address the class imbalance issue, Synthetic Minority Over-sampling Technique (SMOTE) could be implemented to generate synthetic examples of the minority class. This approach would provide a more balanced dataset for training without discarding valuable majority class information.

Enhancing model explainability through SHAP (SHapley Additive exPlanations) values would make complex models like SVM and Random Forest more transparent to healthcare providers. This explainability is critical for clinical adoption, as practitioners need to understand and trust the recommendations provided by ML systems.

The current implementation performs a single train-test split (80/20) with a fixed random state. Future work should implement k-fold cross-validation to provide more robust performance estimates and reduce dependency on a single data partition. This would be particularly valuable given our relatively modest dataset size of 5,110 records.

Additionally, the web application interface could be enhanced to provide not just binary risk assessment but also quantitative risk scores and personalized intervention recommendations based on modifiable risk factors identified in the patient profile.

VI. CONCLUSION

This study demonstrates the efficacy of machine learning techniques in predicting stroke risk using readily available patient data. Our comparative analysis of five algorithms revealed that Support Vector Machine achieved the highest accuracy (93.93%), closely followed by Logistic Regression (93.84%) and Random Forest (93.74%). While all models performed well, the Decision Tree classifier was selected for deployment due to its interpretability in clinical settings despite its slightly lower accuracy (90.51%). Our Flask-based web application successfully bridges the gap between predictive analytics and healthcare practice, offering an intuitive interface that enables healthcare professionals to obtain real-time stroke risk assessments during patient consultations. This approach addresses a critical need in preventive healthcare by providing personalized risk assessment that incorporates both non-modifiable factors (age, gender) and modifiable lifestyle factors (smoking status, BMI) that can guide intervention strategies. The minimal computational requirements of our deployed system make it suitable for resource-constrained healthcare environments, particularly in developing regions where stroke incidence is rising but advanced diagnostic tools remain limited.

There are a number of avenues that need to be explored further in order to fully understand the present study's encouraging findings on the use of machine learning for stroke risk prediction. To improve the model's robustness and generalizability, one important area is to increase the size and diversity of the dataset by adding patient groups from other

healthcare organizations. With the addition of real-time data inputs, the online application might be easily deployed in clinical settings, particularly when linked to EHR systems. Integrating explainable AI (XAI) methods like SHAP or LIME into future endeavors is also important for fostering trust and openness among healthcare providers. Additionally, early warning systems that monitor the risk of stroke over time might be more easily developed with the use of longitudinal patient data. Lastly, a more thorough and precise evaluation of stroke risk might be achieved by expanding the model to include multi-modal data, including medical imaging, genetic profiles, and lifestyle variables.

REFERENCES

- [1] Philip A Wolf, Ralph B D'Agostino, Albert J Belanger, and William B Kannel. Probability of stroke: a risk profile from the framingham study. *Stroke*, 22(3):312–318, 1991.
- [2] Anuj Khosla, Yu Cao, Cliff Chiung-Yu Lin, Hsun-Kai Chiu, Junling Hu, and Honglak Lee. An integrated machine learning approach to stroke prediction. *Proceedings of the 16th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 183–192, 2010.
- [3] Benjamin Letham, Cynthia Rudin, Tyler H McCormick, and David Madigan. Interpretable classifiers using rules and bayesian analysis: Building a better stroke prediction model. *The Annals of Applied Statistics*, 9(3):1350–1371, 2015.
- [4] Samaneh Ghanipour and Yousefzadeh Soroush. Stroke prediction with logistic regression. *ResearchGate*, 2022.
- [5] Laila M Dharamsi, Nida Syed, Anna Lau, Mohammed Farwati, Tina A Tuan, Maan H Khalaf, Amar S Jassar, Michael P Gannon, and Dan T Nguyen. Stroke risk prediction using deep learning: A multi-institutional study. *Frontiers in Neurology*, 12:700358, 2021.
- [6] Yajun Zhang, Xiaoou Yang, and Julie S Ivy. A novel ensemble approach for stroke prediction based on improved gradient boosting and rule ensembles. *Computers & Industrial Engineering*, 132:81–90, 2019.
- [7] Lanhao Chen, Paul Bentley, and Daniel Rueckert. Machine learning in acute stroke neuroimaging. *Frontiers in Neurology*, 11:369, 2020.
- [8] Gustavo Saposnik et al. Iscore: a risk score to predict death early after hospitalization for an acute ischemic stroke. *Circulation*, 123(7):739–749, 2011.
- [9] Larry B Goldstein, Cheryl D Bushnell, Robert J Adams, Lawrence J Appel, Lynne T Braun, Seemant Chaturvedi, Mark A Creager, Antonio Culebras, Robert H Eckel, Robert G Hart, et al. A comprehensive review of inflammatory biomarkers in ischemic stroke and their utility in prediction and prognosis. *Journal of Stroke and Cerebrovascular Diseases*, 26(12):2712–2724, 2017.
- [10] Karen C Johnston, Askiel Bruno, Qi Pauls, Christiana E Hall, Kevin M Barrett, William Barsan, Amy Fansler, Kristen Van de Bruinhorst, Scott Janis, and Valerie L Durkalski-Mauldin. Blood biomarkers of ischemic stroke and their role in stroke prediction and diagnosis. *Neurotherapeutics*, 16(3):584–592, 2019.
- [11] Nicole Neuhaus, Ira S Hofer, Azra Bihorac, Tezcan Ozrazgat-Baslanti, Xiaolin Jia, Ashkan Ebadi, Anton Oberhofer, and Mengye Li. Application of federated learning for stroke prediction in multi-institutional healthcare systems. *Journal of the American Medical Informatics Association*, 28(12):2651–2659, 2021.
- [12] Sharif Mahmood, John P Greenwood, Chris P Gale, Paul D Baxter, Rizwan Kamran Shoukat, Alistair S Hall, Syed Altaf Hussain, John Deanfield, and Stephen B Wheatcroft. Temporal patterns of biomarker levels improve stroke prediction beyond single measurements. *European Heart Journal*, 43(34):3234–3244, 2022.
- [13] Gladys Mirella Montoro Silva, Tiago Martins Valente, Walmir Xavier Silva, Diego Brito, and André Carlos Ponce Leon Ferreira de Carvalho. Feature selection for stroke prediction using statistical learning methods. *Journal of Intelligent & Fuzzy Systems*, 35(4):4747–4755, 2018.
- [14] Jung Hyun Lee, Youngjin Kim, Yeon-Joo Kim, Eunjeong Park, Joonho Choi, Hee-Jin Lee, Ji Hoe Heo, and Hyo Suk Nam. Explainable ai for clinical decision support: contextualizing machine learning predictions of stroke risk. *Journal of Medical Systems*, 45:1–13, 2021.

- [15] David Rodriguez-Luna, Brett Cucchiara, Manuel Gómez-Choco, Alejandro Tomasello, Pere Cardona, Marc Roqué, Xabier Urrea, Víctor Obach, Álvaro Cervera, María Hernández-Pérez, et al. Clinical impact of artificial intelligence systems in stroke prediction: A prospective evaluation. *Stroke*, 53(5):1658–1666, 2022.
- [16] Anil K Verma, Rahul Garg, Uma Sundar, Jaideep Gogtay, Urmila Thatte, Abhijit Raut, S Cynthia, and Sapna E Sridharan. Barriers to adoption of machine learning models for clinical stroke prediction. *Neurology India*, 69(3):653–659, 2021.
- [17] Junyu Chang, Yi Zhang, Jia Ren, Yuan Wang, Jianfeng Feng, Dan Fan, and Liang Yu. Systematic review of machine learning models for stroke prediction. *Frontiers in Neurology*, 13:852139, 2022.
- [18] Daniel Oliveira, Marcel Santos, Carolina Azevedo, Rodrigo Moura, Nilson Santos, Rafael Pereira, and Alice Ribeiro. Machine learning for stroke prediction: Current approaches and future directions. *Biomedical Signal Processing and Control*, 79:104091, 2023.
- [19] Andrea Morotti, Grégoire Boulouis, Andreas Charidimou, Kristin Schwab, Christina Kourkoulis, Christopher D Anderson, M Edip Gurol, Anand Viswanathan, Steven M Greenberg, Jonathan Rosand, et al. Machine learning for outcome prediction in subarachnoid hemorrhage. *International Journal of Stroke*, 15(1):96–98, 2020.
- [20] Masahito Katsuki, Naoyuki Narita, Naoya Ishida, Osamu Watanabe, Siqi Cai, Fumiaki Koh, and Teiji Tominaga. Machine learning analysis to identify stroke predictors and their interactions: A prospective cohort study. *BMJ Open*, 11(5):e046137, 2021.
- [21] Nazia Kamal, Ashfaq Shuaib, Eric E Smith, Thomas Jeerakathil, and Michael D Hill. The future of stroke risk prediction: Moving from identification to implementation. *International Journal of Stroke*, 17(1):13–20, 2022.